

# MAC Flooding attack and mitigation

## PART1

### Summary

This report analyses a Layer 2 MAC Flooding attack using Kali Linux and Wireshark. It demonstrates mitigating the threat in Cisco Packet Tracer by implementing Port Security to automatically shut down unauthorized switch ports.

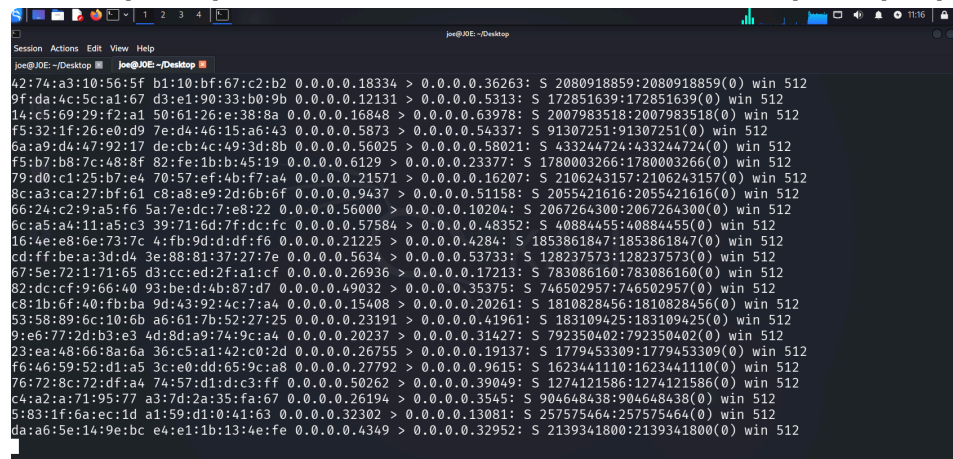
### Evidence

## Part 1: Attack Execution & Traffic Analysis (Kali Linux & Wireshark)

### Step 1: Executing the Attack and Capturing Traffic

- **Objective:** Overflow the Switch CAM Table using synthetic frames and log the generated network traffic.
- **Execution:**
  1. Initiated the MAC flooding attack using `macof` on interface `eth0` : **`sudo macof -i eth0`**
  2. Simultaneously recorded 20,000 frames using `tcpdump` to a PCAP capture file:

**`sudo tcpdump -i eth0 -U -c 20000 -w ~/mac-flood-capture.pcap`**

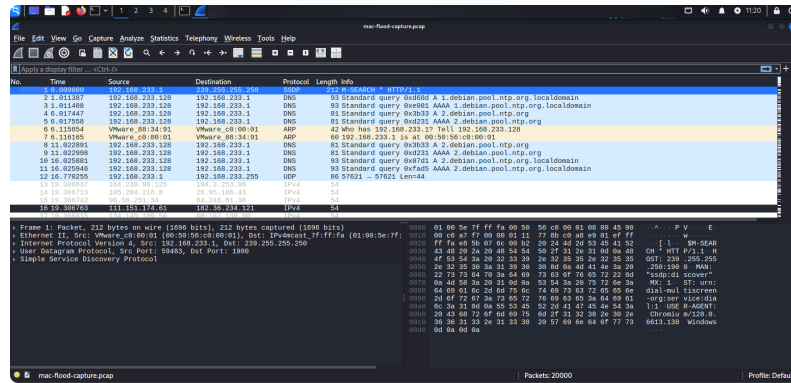


```
42:74:a3:10:56:5f b1:10:bf:67:c2:b2 0.0.0.0.18334 > 0.0.0.0.36263: S 2080918859:2080918859(0) win 512
9f:da:4c:5c:a1:67 d3:e1:90:33:b0:9b 0.0.0.0.12131 > 0.0.0.0.5313: S 172851639:172851639(0) win 512
14:c5:69:29:f2:a1 50:61:26:e:38:8a 0.0.0.0.16848 > 0.0.0.0.63978: S 2007983518:2007983518(0) win 512
f5:32:1f:26:e0:d9 7e:d4:46:15:a6:43 0.0.0.0.5873 > 0.0.0.0.54337: S 91307251:91307251(0) win 512
6a:a9:d4:47:92:17 de:cb:4c:49:3d:8b 0.0.0.0.56025 > 0.0.0.0.58021: S 433244724:433244724(0) win 512
f5:b7:b8:7c:48:8f 82:fe:1b:b:45:19 0.0.0.0.6129 > 0.0.0.0.23377: S 1780003266:1780003266(0) win 512
79:d0:c1:25:b7:e4 70:57:ef:4b:f7:a4 0.0.0.0.21571 > 0.0.0.0.16207: S 2106243157:2106243157(0) win 512
8c:a3:ca:27:bf:61 c8:a8:e9:2d:6b:6f 0.0.0.0.9437 > 0.0.0.0.51158: S 2055421616:2055421616(0) win 512
66:24:c2:9:a5:f6 5a:7e:dc:7:e8:22 0.0.0.0.56000 > 0.0.0.0.10204: S 2067264300:2067264300(0) win 512
6c:a5:a4:11:a5:c3 39:71:6d:7f:dc:fc 0.0.0.0.57584 > 0.0.0.0.48352: S 40884455:40884455(0) win 512
16:4e:e8:6e:73:7c 4:fb:9d:d:df:f6 0.0.0.0.21225 > 0.0.0.0.4284: S 1853861847:1853861847(0) win 512
cd:ff:be:a:3d:d4 3e:88:81:37:27:7e 0.0.0.0.5634 > 0.0.0.0.53733: S 128237573:128237573(0) win 512
67:5e:72:1:71:65 d3:cc:ed:2f:a1:cf 0.0.0.0.26936 > 0.0.0.0.17213: S 783086160:783086160(0) win 512
82:dc:cf:9:66:40 93:be:d:4b:87:d7 0.0.0.0.49032 > 0.0.0.0.35375: S 746502957:746502957(0) win 512
c8:1b:6f:40:fb:ba 9d:43:92:4c:7:a4 0.0.0.0.15408 > 0.0.0.0.20261: S 1810828456:1810828456(0) win 512
53:58:89:6c:10:6b a6:61:7b:52:27:25 0.0.0.0.23191 > 0.0.0.0.41961: S 183109425:183109425(0) win 512
9:e6:77:2d:b3:e3 4d:8d:a9:74:9c:a4 0.0.0.0.20237 > 0.0.0.0.31427: S 792350402:792350402(0) win 512
23:ea:48:66:8a:6a 36:c5:a1:42:c0:2d 0.0.0.0.26755 > 0.0.0.0.19137: S 1779453309:1779453309(0) win 512
76:46:59:52:d1:a5 3c:e0:dd:65:9c:a8 0.0.0.0.27792 > 0.0.0.0.9615: S 1623441110:1623441110(0) win 512
76:72:8c:72:df:a4 74:57:d1:d:c3:ff 0.0.0.0.50262 > 0.0.0.0.39049: S 1274121586:1274121586(0) win 512
c4:a2:a:71:95:77 a3:7d:2a:35:fa:67 0.0.0.0.26194 > 0.0.0.0.3545: S 904648438:904648438(0) win 512
5:83:1f:6a:ec:1d a1:59:d1:0:41:63 0.0.0.0.32302 > 0.0.0.0.13081: S 257575464:257575464(0) win 512
da:a6:5e:14:9e:bc e4:e1:1b:13:4e:fe 0.0.0.0.4349 > 0.0.0.0.32952: S 2139341800:2139341800(0) win 512
```

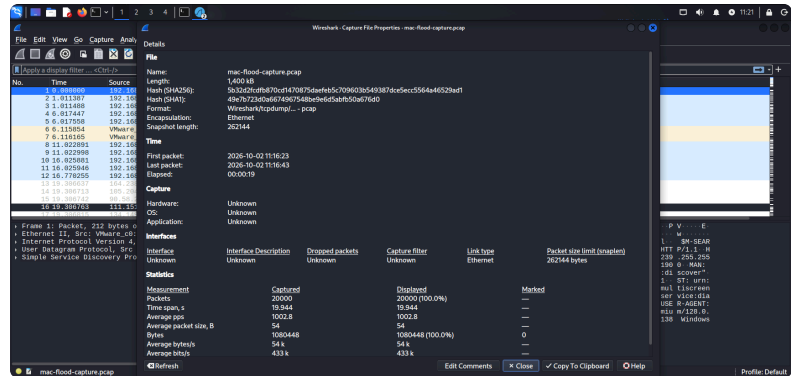
### Step 2: Traffic Analysis in Wireshark

- **Objective:** Inspect packet properties and evaluate the volume of generated MAC addresses.
- **Analysis:**

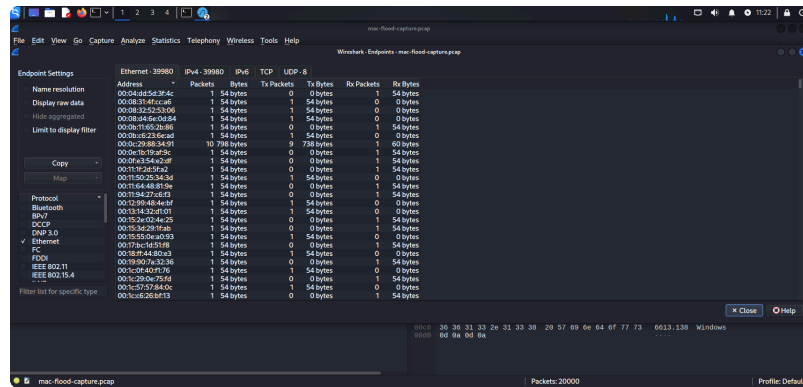
## 1. Opened mac-flood-capture.pcap in Wireshark.



## 2. Navigated to **Statistics** → **Capture File Properties** to review the total traffic volume and the short duration in which the 20,000 packets were generated.



## 3. Navigated to **Statistics** → **Endpoints** and inspected the **Ethernet** tab.



- **Observation:** Thousands of unique, randomized source MAC addresses were observed, confirming an active MAC Flooding (CAM Table Overflow) attack designed to force the switch into fail-open (hub-like) mode.

# PART2

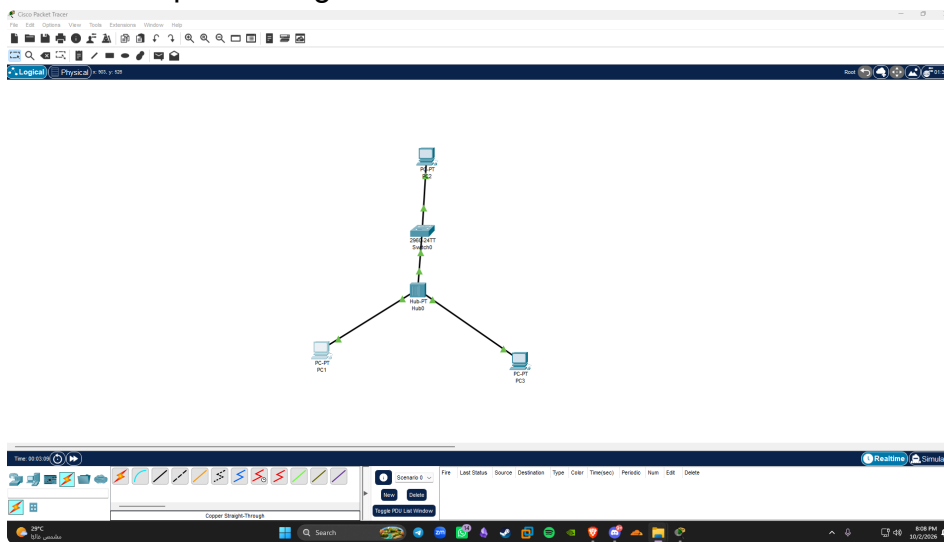
## Step 1: Topology Setup & Initial State

- **Topology:**
  - **Switch:** SW1 (Cisco Switch)
  - **Legitimate Hosts:**
    - PC1 : 10.10.10.10

- PC2 : 10.10.10.11
- **Rogue Host:**
  - PC3 : 10.10.10.12
  - **Populating the CAM Table:** On PC1 , sent ICMP traffic to PC2 : **ping 10.10.10.11**
  - Verified SW1 CAM Table: **show mac address-table**

## Step 2: Demonstrating Rogue Device Spoofing

- Connect PC3 (Rogue PC) to interface Fa0/1 (previously used by PC1 ) and send ICMP traffic: **ping 10.10.10.11**
- On SW1, execute: **show mac address-table**
- **Observation:** SW1 learned the rogue MAC address on Fa0/1 without restrictions, proving that default port settings are vulnerable.



## Step 3: Hardening the Switch with Port Security

To secure interface FastEthernet0/1 , the dynamic MAC address table was cleared and Port Security policies were applied:

**clear mac address-table dynamic**

**configure terminal**

**interface fastEthernet0/1**

**switchport mode access**

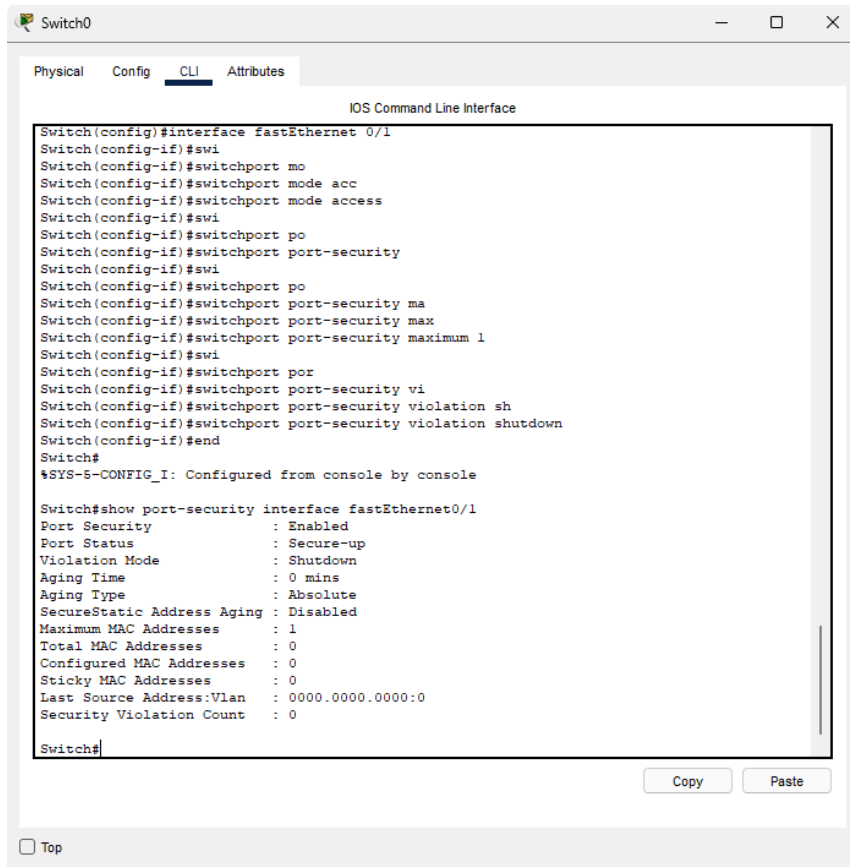
**switchport port-security**

**switchport port-security maximum 1**

**switchport port-security violation shutdown**

**end**

## show port-security interface fastEthernet0/1



The screenshot shows a network switch configuration window titled "Switch0". It has tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The configuration commands for interface fastEthernet0/1 are as follows:

```
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#swi
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#swi
Switch(config-if)#switchport po
Switch(config-if)#switchport port-security
Switch(config-if)#swi
Switch(config-if)#switchport po
Switch(config-if)#switchport port-security ma
Switch(config-if)#switchport port-security max
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#swi
Switch(config-if)#switchport por
Switch(config-if)#switchport port-security vi
Switch(config-if)#switchport port-security violation sh
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#end
Switch#
```

Below the configuration commands, the status of the port security is displayed:

```
#SYS-5-CONFIG_I: Configured from console by console

Switch#show port-security interface fastEthernet0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0

Switch#
```

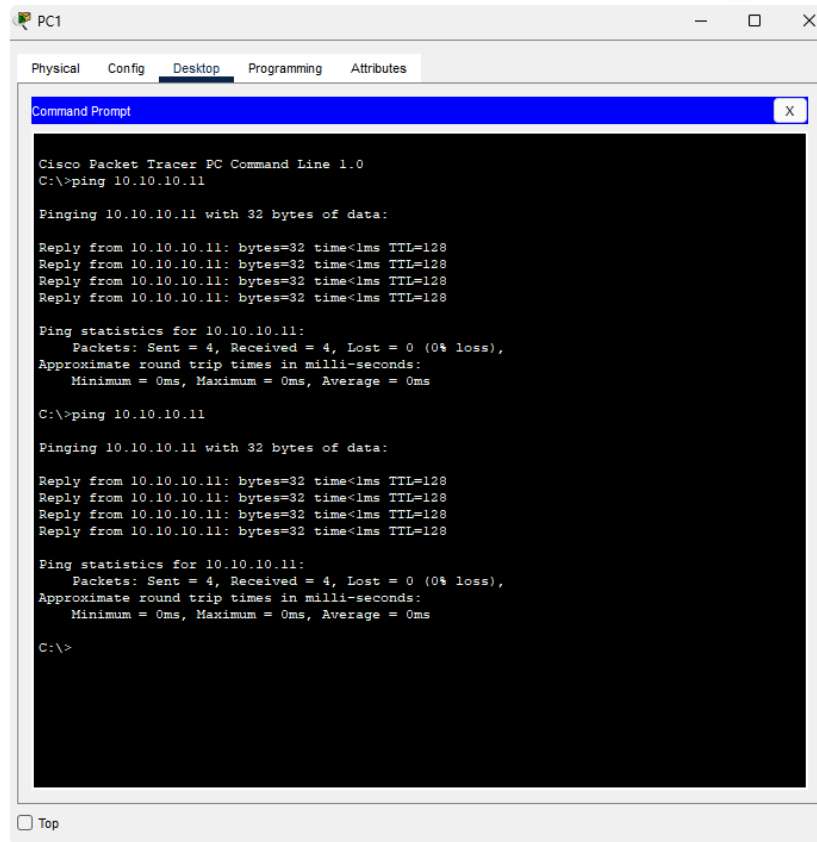
At the bottom of the CLI window, there are "Copy" and "Paste" buttons. Below the CLI window, there is a "Top" button.

## Step 4: Testing & Violation Trigger

### 1. Valid Device Traffic:

Ran ping 10.10.10.11 from PC1 .

- Verification: Port status displayed **Secure-up** with 1 authorized MAC address logged.



The screenshot shows a Cisco Packet Tracer PC Command Prompt window for PC1. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt displays the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.11

Pinging 10.10.10.11 with 32 bytes of data:

Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.10.11

Pinging 10.10.10.11 with 32 bytes of data:

Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
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Ping statistics for 10.10.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

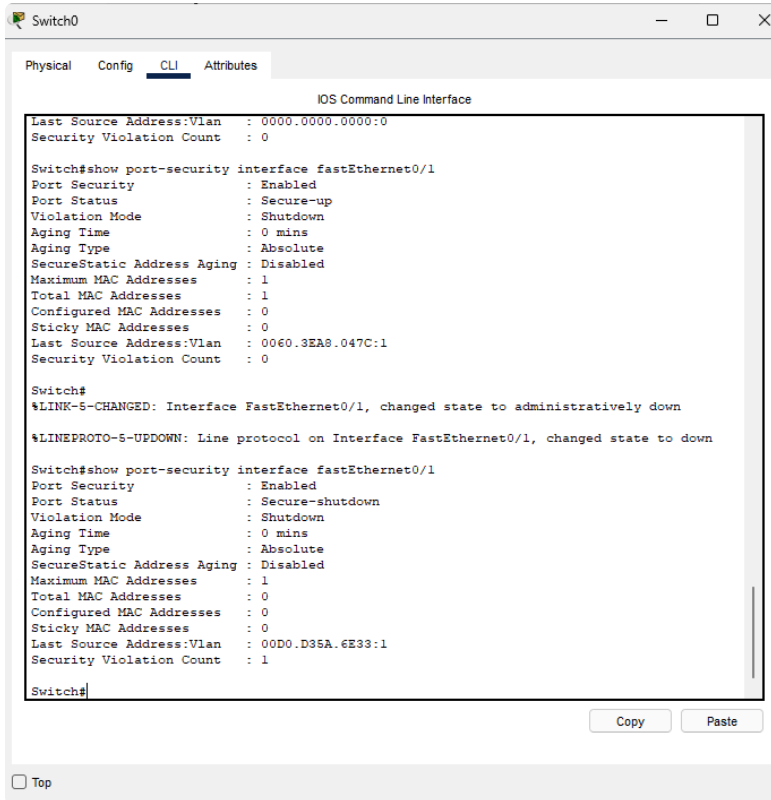
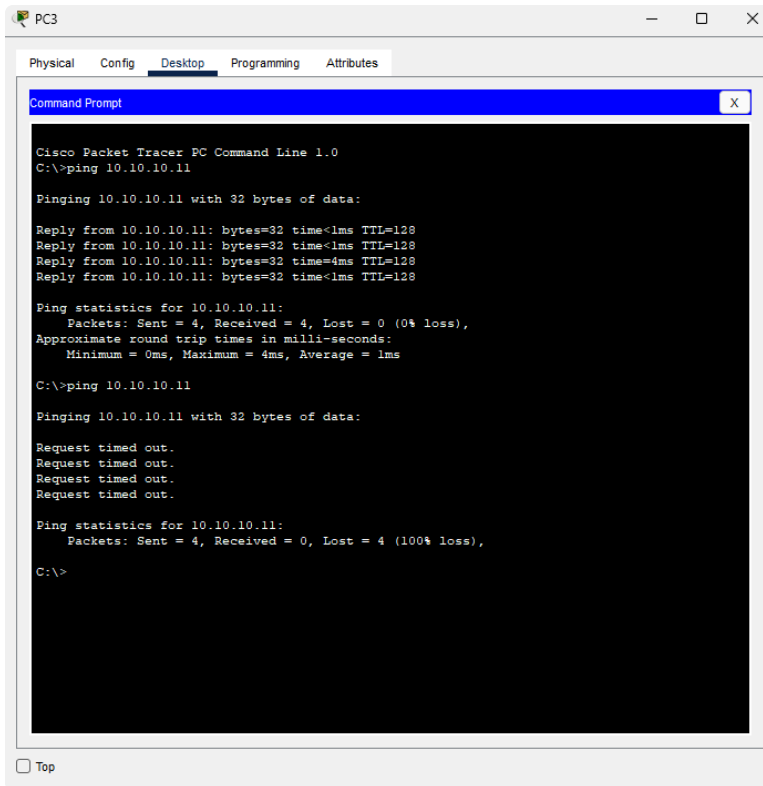
C:\>
```

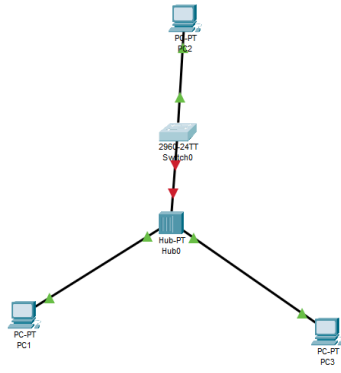
At the bottom left of the window, there is a checkbox labeled "Top" which is currently unchecked.

## 2. Unauthorized Device Traffic:

Connected PC3 to Fa0/1 and attempted ping 10.10.10.11 .

- Verification: Executed show port-security interface fastEthernet0/1 on SW1.
- **Result:** Port status changed to **Secure-shutdown** , effectively blocking unauthorized access and neutralizing MAC flooding/spoofing attempts.

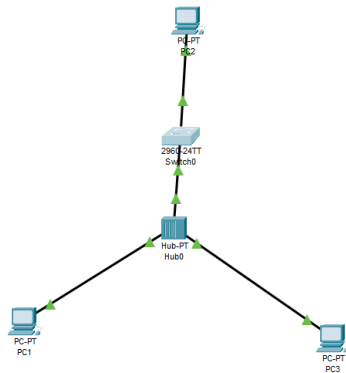
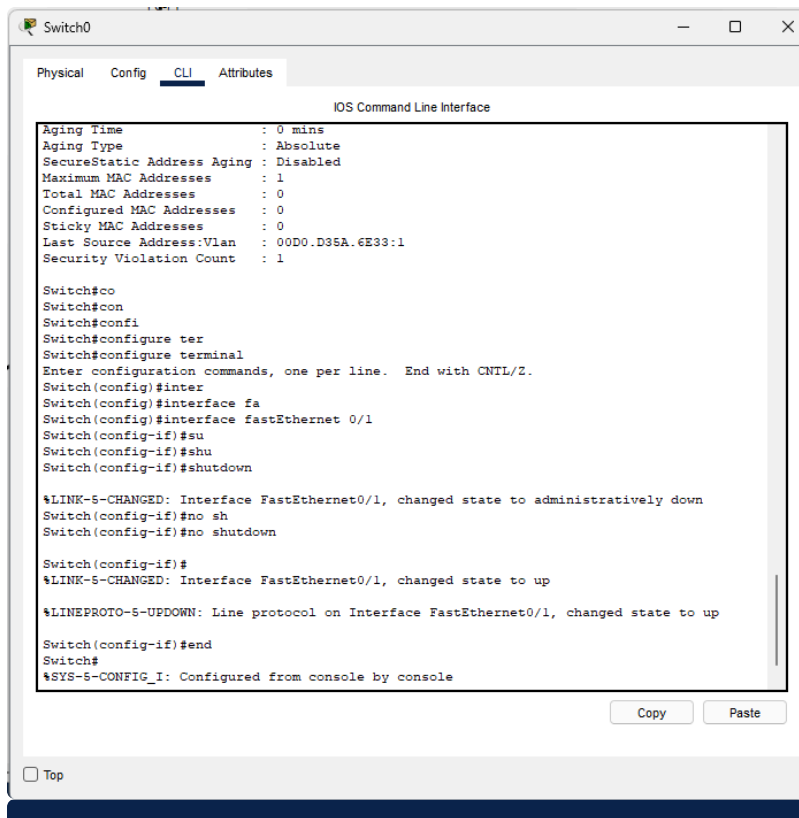




## Step 5: Port Recovery Procedure

```
configure terminal
interface fastEthernet0/1
shutdown
no shutdown
end
```

**Validation:** Verified connectivity by issuing `ping 10.10.10.11` from PC1 . Ping succeeded and port status returned to normal operational status.



## Recommendation

- **Enable Sticky MAC Learning:** Use switchport port-security mac-address sticky to dynamically learn and write authorized MAC addresses to the running configuration.
- **Disable Unused Ports:** Shut down all unused switch ports manually ( shutdown ) and place them in an isolated VLAN.
- **Implement DHCP Snooping & DAI:** Combine Port Security with DHCP Snooping and Dynamic ARP Inspection (DAI) for comprehensive Layer 2 defense.